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Final Exam – June 8, 2026

Closed-book exam. No notes, no laptop, no phone, no internet, no AI assistance, no anything. Just your brain and a pen!

Duration: 1 hour 30 minutes

Total number of points: 100

Instructions

Questions are grouped into logical sections to ease your thought process. There are two types of questions:

- **Multiple-choice:** One correct choice per question. Please circle the correct answer.
- **Free-form:** Short and concise answers.

In all cases, there is no penalty for wrong answers.

I. Image generation architectures (25 points)

1. (1 point) A good image-generation backbone should be able to:
 - A. Prioritize local texture while ignoring long-range structure.
 - B. Use only global interactions while discarding fine spatial details.
 - C. Accept conditions only after the latent has already been finalized.
 - D. Capture global structure, preserve local details, use conditions, and scale.
2. (1 point) In a U-Net, skip connections are especially useful because they:
 - A. Preserve local information from the downsampling path.
 - B. Increase patch-token count before the attention blocks.
 - C. Remove the need for any upsampling operation.
 - D. Collapse all spatial features into one bottleneck vector.

3. (1 point) In adaLN-style conditioning, suppose a token representation is updated as

$$x \leftarrow x + \alpha \star \text{Operation}(\text{LN}(x) \star (1 + \gamma) + \beta).$$

Which statement matches the roles of α , γ , and β ?

- A. α shifts $\text{LN}(x)$, γ gates the residual update, and β changes patch size.
 - B. α changes the sequence length, γ shifts the token, and β chooses the condition.
 - C. α gates the update, γ scales the normalized token, and β shifts it.
 - D. α removes the residual path, γ performs pooling, and β unpatchifies tokens.
4. (2 points) A Diffusion Transformer first adapts an image or latent by:
- A. Treating the entire latent as one scalar token.
 - B. Dividing it into patches and embedding those patches as tokens.
 - C. Applying convolutional downsampling without tokenization.
 - D. Projecting only condition embeddings and ignoring image patches.
5. (2 points) Adaptive layer normalization in DiT is used to:
- A. Modulate token features with quantities derived from conditions.
 - B. Change patch size to control the number of latent tokens.
 - C. Provide absolute coordinates as extra image channels.
 - D. Concatenate U-Net skip features in the decoder path.
6. (2 points) In DiT scaling, using a smaller patch size usually:
- A. Decreases the token count while keeping the same latent size.
 - B. Changes only the embedding width, not the sequence length.
 - C. Increases FLOPs because there are more tokens.
 - D. Removes the need to encode positions in the attention stack.
7. (2 points) The motivation for joint attention in MM-DiT is to:
- A. Prevent text and image representations from ever interacting.
 - B. Replace conditioning with a single global class label.
 - C. Keep each modality in isolated streams to prevent information leakage.
 - D. Let image and condition tokens interact more flexibly.
8. (1 point) Rotary position embeddings are used to:
- A. Encode absolute position information as input.
 - B. Encode relative position information inside attention.
 - C. Gate normalized token features with timestep embeddings.
 - D. Copy high-resolution U-Net features into decoder layers.

9. (3+4 points) **U-Net versus DiT.** (i) State one architectural advantage of U-Nets for image generation. (ii) State one reason Transformers became attractive for modern image generation.

(i)

(ii)

10. (3+3 points) **Conditioning in DiT and MM-DiT.** (i) Explain the role of adaLN-style modulation. (ii) Why might joint attention be preferred for richer text/image conditioning?

(i)

(ii)

II. Model training (25 points)

1. (1 point) A modern image-generation training lifecycle is best summarized as:
 - A. Pre-train, distill, then post-training and tuning.
 - B. Tune first on a few subjects, then pre-train broadly.
 - C. Pre-train, post-train, optionally tune, and distill.
 - D. Distill first, then pre-train, post-train, and tune.
2. (1 point) The motivation for changing the timestep sampling distribution during training is to:
 - A. Emphasize hard timesteps during training.
 - B. Ensure that only nearly clean examples appear during training.
 - C. Ensure that all timesteps are sampled equally often.
 - D. Emphasize easy timesteps during training.

3. (1 point) REPA is an example of adding a loss that:
 - A. Changes timestep sampling to oversample easy examples.
 - B. Adds preference rewards directly to final samples.
 - C. Distills many teacher steps into one student step.
 - D. Aligns model features with a pretrained image representation.
4. (2 points) Curriculum learning for image generation typically moves from:
 - A. Long prompts and high resolution to short prompts and low resolution.
 - B. Easier settings to harder settings as training progresses.
 - C. Randomly mixed difficulty to a staged easy-only dataset.
 - D. Fixed resolution and prompt complexity throughout training.
5. (2 points) Preference tuning methods such as ReFL, Flow-GRPO, and Diffusion-DPO aim to:
 - A. Use human or reward-model preferences to steer generation.
 - B. Use the image generation model to tune human preferences.
 - C. Prefer a single subject identity via the use of a rare token.
 - D. Compress a many-step sampler into a preferred, smaller-step student model.
6. (2 points) DreamBooth-style tuning is mainly used to:
 - A. Improve general quality through broad continued training.
 - B. Train every parameter from scratch on a large generic dataset.
 - C. Bind a few subject images to a rare token.
 - D. Speed up sampling by matching teacher transitions.
7. (2 points) LoRA is useful for tuning because it:
 - A. Requires all original model weights to be retrained fully for maximum quality.
 - B. Adds full-rank updates to every layer in the backbone of the model.
 - C. Changes prompt wording at inference time without the need for explicit re-training.
 - D. Trains small low-rank adapters instead of all weights.
8. (1 point) In image generation, distillation most often tries to:
 - A. Increase the number of sampling steps for higher fidelity.
 - B. Produce good samples with fewer sampling steps.
 - C. Specialize a model to one specific subtask.
 - D. Train a smaller student model on the larger teacher's outputs, using the same number of sampling steps.

9. (3+4 points) **Training lifecycle.** (i) Name the four broad stages of a modern image-generation training lifecycle. (ii) Give the high-level purpose of each stage.

(i)

(ii)

10. (3+3 points) **Distillation.** (i) Why is teacher/student distillation attractive for diffusion or flow models? (ii) Name one limitation or trade-off of an approach such as progressive distillation, consistency models, DMD, or adversarial distillation.

(i)

(ii)

III. Evaluation (25 points)

1. (1 point) A generated image can be visually realistic while ignoring the prompt, or follow the prompt while looking poor. This motivates evaluating:
 - A. Diversity and safety.
 - B. Speed and memory use.
 - C. Aesthetics and prompt adherence.
 - D. Text rendering and editing fidelity.
2. (1 point) Pairwise human comparisons are often easier than absolute ratings because they:
 - A. Ask raters which of two images is better.
 - B. Ask raters to explicitly calibrate a five-point scale that can then be used to rank all images.
 - C. Require each image to match one reference exactly.
 - D. Produce single-sided evaluations that are easy for diagnostics.

3. (1 point) Elo-style ratings improve on raw win rate by:
 - A. Averaging binary pass rates through an easy-to-compute formula.
 - B. Counting only wins against the same fixed opponent.
 - C. Treating every opponent as equally informative.
 - D. Accounting for the strength of the opponent.
4. (2 points) FID evaluates generated images by:
 - A. Comparing raw pixel distances to a reference image.
 - B. Comparing real and generated feature distributions.
 - C. Asking a prompt-adherence judge yes/no questions.
 - D. Averaging human pairwise preference ratings.
5. (2 points) A CLIPScore-style metric may give similar scores for an image paired with the prompts “A cute teddy bear is reading a book” and “A cute book is reading a teddy bear,” even though the relation is reversed. What limitation does this illustrate?
 - A. Reference-free metrics require a ground-truth image before they can be computed.
 - B. Feature-distribution metrics cannot estimate whether images look realistic.
 - C. Global image-text alignment can miss compositional or relational swaps.
 - D. Pairwise human ratings cannot compare two images generated from related prompts.
6. (2 points) Suppose an edited image preserves the right object and style, but the object is shifted a few pixels relative to the reference. Which evaluation concern applies?
 - A. Pixel-wise metrics such as MSE can over-penalize small spatial shifts.
 - B. Pairwise human ratings cannot compare two images from the same prompt.
 - C. Reference-free metrics require the shifted reference image.
 - D. Elo-style ratings become exact only when the shift is small enough.
7. (2 points) A best practice for MLLM-as-a-Judge evaluation is to:
 - A. Avoid tuning it with human preferences to avoid introducing bias.
 - B. Keep A/B ordering fixed across pairwise comparisons for determinism.
 - C. Ask for the final score before the evidence.
 - D. Ask for the evidence before the final score.

8. (1 point) A prompt says, “A cute teddy bear is reading a book.” Suppose an MLLM-as-a-Judge first asks whether there is a teddy bear, whether there is a book, and whether the teddy bear is reading the book. This decomposition is useful because it:
- A. Uses the same grading rubric across prompts.
 - B. Makes it easier to localize failures of specific claims.
 - C. Ensures that the final score is unbiased.
 - D. Removes the need to inspect the generated image.
9. (2+2+3 points) **Human evaluation.** (i) What signal does a pointwise rating record? (ii) What signal does a pairwise rating record? (iii) Why can Elo-style updates be more informative than raw win rate?

- (i)
- (ii)
- (iii)

10. (3+3 points) **Metrics and judges.** (i) Why are fixed metrics such as FID, CLIPScore, SSIM, or LPIPS useful but incomplete? (ii) What is the role of MLLM-as-a-Judge methods?

- (i)
- (ii)

IV. Trending topics (25 points)

1. (1 point) Many strong open-weight text-to-image generators share the following high-level design pattern:
- A. Pixel-space U-Nets with no latent bottleneck or text encoder.
 - B. VAE latents with convolutional blocks and text embeddings trained from scratch.
 - C. Flow training with VAE latents but no multimodal conditioning.
 - D. Flow training, MM-DiT, VAE latents, and strong text embeddings.

2. (1 point) Suppose an open-weight text-to-image model performs close to leading closed models on standard image-generation evaluations. Why is this significant?
 - A. It proves open-weight models are safer than closed models.
 - B. Strong generation capabilities become easier to inspect, adapt, and deploy outside a few labs.
 - C. It means evaluation is no longer needed once model weights are public.
 - D. It shows that video and image generation now have identical difficulty.
3. (1 point) A core added difficulty in moving from image generation to video generation is:
 - A. Handling time, temporal consistency, and larger compute.
 - B. We now need a time tokenizer to encode the time dimension.
 - C. The lack of spatial structure caused by multiple frames.
 - D. Evaluating one static frame with image-only metrics.
4. (2 points) Fréchet Video Distance is used as:
 - A. A spatio-temporal VAE architecture choice for compression.
 - B. A method for enforcing frame consistency during sampling.
 - C. A video evaluation metric analogous in spirit to FID.
 - D. A prompt format for describing camera motion and duration.
5. (2 points) A guardrailed image-editing pipeline can avoid full regeneration by:
 - A. Regenerating the full image from the edited prompt.
 - B. Ranking fully regenerated candidates by aesthetic score.
 - C. Ignoring the source image and generating from text only.
 - D. Using a VLM to choose actions for editing software.
6. (2 points) For diffusion language models, a practical analog of image noise is:
 - A. Gaussian noise added to discrete token identities.
 - B. Masking a subset of discrete tokens.
 - C. Dropping punctuation while keeping all semantic tokens.
 - D. Shuffling all tokens while treating order as unchanged.

7. (2 points) Diffusion LLM inference is **best** described as:
- A. Starting masked, then predicting, committing, and remasking tokens.
 - B. Predicting tokens one by one from left to right.
 - C. Filling every mask in one pass and discarding low-confidence tokens from the final sequence.
 - D. Applying Gaussian noise to token representations and decoding once at the end.
8. (1 point) In fill-in-the-middle code completion, a model sees a prefix and suffix and fills the missing span. Why is this natural for diffusion-style language models?
- A. A left-to-right model can use the suffix without any special setup.
 - B. Each missing token can be filled independently in one pass.
 - C. The missing span can be masked, then predicted and revised using both sides as context.
 - D. Training uses next-token likelihood only, so masking is unnecessary.
9. (3+4 points) **Images, videos, and editing.** (i) What changes when extending image generation to video generation? (ii) Why might an editing-action pipeline be safer than fully regenerating an edited image?

(i)

(ii)

10. (3+3 points) **Diffusion for text.** (i) Contrast autoregressive LLM generation with diffusion-style LLM generation. (ii) Name one advantage and one challenge of diffusion LLMs.

(i)

(ii)

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We hope you enjoyed spending this quarter with us. Happy graduation if this was your last quarter. Wishing you a relaxing summer and the best of success ahead!